What Conformal Gravity Must Assume
to Explain Galactic Rotation Curves

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Abstract

Conformal (Weyl) gravity admits a static spherically symmetric vacuum solution whose metric function carries a term linear in r. That term is the basis of a long-standing proposal to fit galactic rotation curves without dark matter. We ask a reverse-physics question about it: not whether the fits succeed, but *what must be assumed for them to mean anything.*

Eight results, each an exact symbolic computation with machine-checked controls. (i) The linear term is not an ansatz: in the conformal gauge the Bach vacuum equation is exactly the *biharmonic* equation $\nabla^4 B = 0$, so γr is the point-source response of the governing operator, on the same footing as the Newtonian $1/r$. (ii) If the baryonic Tully–Fisher relation holds, γ is forced to be a universal constant and is identified with the MOND acceleration scale, $\gamma = a_0/2c^2$; a γ carrying a mass-proportional piece gives the wrong slope. (iii) Conformal invariance forces the source to be trace-free, so a perfect fluid must satisfy $p = \rho/3$ and pressureless baryons are excluded outright. (iv) The standard repair—breaking the symmetry with a conformally coupled scalar—fails in its simplest form: a *constant* vacuum expectation value pins the Ricci scalar and thereby forces $\gamma = 0$, removing the very term at issue.

Four further results concern the surviving branch, and *none of them is new physics*; they are computed here with the ansatz removed, and the literature that already contains them is identified in §9. (v) γ is not a conformal invariant: the conformal group acts on the four parameters of the general solution through a single Möbius map of the radius, and shifts γ. (vi) The frame in which the scalar is constant is reached by exactly that map, and a mass *generated* by the scalar makes the scalar the conformal factor of the metric the particle moves in, so such a particle follows geodesics of the frame with $\gamma = 0$. (vii) In that frame the pair is an exact solution: Bach flatness and the Einstein condition hold together, the geometry is Schwarzschild–de Sitter, and u is a conformal invariant while γ is not. (viii) Constructing the Green's function of the biharmonic operator, γ is the second moment of the source—the total mass—while the Newtonian coefficient is the fourth moment, tracking $M\langle r^2\rangle$.

Results (vi) and (vii) are Hobson and Lasenby's [10], who reach the stronger conclusion that the resulting rotation curves have no flat region at all; result (viii) is Mannheim and Kazanas' [5] and is the substance of the standing objection to the Newtonian limit. We reproduce them under machine-checked controls and do not claim them.

What the assembly gives is a map of the assumption rather than a new result. The linear coefficient decomposes as $\gamma = \gamma_0 + \gamma_*(M)$—a boundary datum plus a sourced piece—which is the parametrisation the rotation-curve fits already use [6]. Our result (ii) says a mass-proportional piece moves the Tully–Fisher slope, while those fits keep both terms and succeed. An earlier draft recorded that as an unresolved tension; it is not one, and the resolution is already published [12]. Their eq. (7) gives $v^4 = (BM/M_\odot)(1 + N^*/D)$ with $B/M_\odot = 2c^2G\gamma_0$ and $D = \gamma_0/\gamma^*$, so

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the mass-proportional piece enters exactly as we say—and the fits sit where $N^* \lesssim D$, so the correction factor lies between 1 and 2 and the slope is near 4 without being 4. Our conditional and their fit agree; what we add is that the slope is *exactly* 4 only in the limit where the sourced piece vanishes.

1 The question

Conformal gravity has been proposed as an explanation of galactic rotation curves without dark matter, on the strength of a term linear in r in its static spherically symmetric vacuum solution. The literature on whether the fits succeed is substantial and we add nothing to it.

We ask instead the reverse-physics question: *what has to be true for the explanation to be an explanation?* A fit with a per-galaxy free parameter is a description; a fit with a parameter forced by the theory is a prediction. Which of these the proposal is depends on facts that can be settled by computation, and this paper settles four of them.

Every statement below is an exact symbolic identity over $\mathbb{Q}$, carried out with unspecified functions rather than sampled fixtures, with controls designed to fail if the claim were vacuous. Certificates and verifier commands are listed in §10.

Conventions. We work in $D = 4$, signature $(-,+,+,+)$, with the static spherically symmetric ansatz

$$\mathrm{d}s^2 \;=\; -B(r)\,\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{B(r)} + r^2\mathrm{d}\Omega^2,$$

the conformal gauge $g_{tt}g_{rr} = -1$. §6 addresses what that gauge costs. B_{ab} denotes the Bach tensor.

2 The linear term is forced, and it is a point-source response

Theorem 2.1 (Completeness without an ansatz)**.** *Let B be any C^4 function, nonvanishing on an interval. Then $B_{ab} = 0$ identically if and only if*

$$(rB)'''' = 0 \qquad \textit{and} \qquad c_1^2 - 3c_0c_2 = 1,$$

where $B = c_0/r + c_1 + c_2 r + c_3 r^2$.

The proof is a decomposition. With $L := (rB)''''$ and N the nonlinear remainder, the three independent Bach rows satisfy, as exact identities in B and its derivatives,

$$B_{\theta\theta} = \frac{N}{24r^2}, \qquad B_{tt} = \frac{B\,(N + 2r^3BL)}{24r^4}, \qquad B_{rr} = \frac{-N + 2r^3BL}{24r^4B},$$

so all three lie in the span of $\{N,\ r^3BL\}$, and both generators are recoverable from the rows. Hence $B_{ab} = 0$ forces $L = 0$ and $N = 0$.

The essential point is that L is *linear*. Substituting $u = rB$ turns $L = 0$ into $u'''' = 0$, whose solution space is the cubics by four integrations—no ansatz. On that space N collapses to the constant $4(1 + 3c_0c_2 - c_1^2)$, leaving one algebraic condition.

Theorem 2.2 (The vacuum equation is biharmonic)**.** *In spherical symmetry $\nabla^2 f = f'' + 2f'/r$, and for unspecified B*

$$\nabla^4 B \;=\; B'''' + \frac{4B'''}{r} \;=\; \frac{1}{r}\,(rB)''''.$$

So Theorem 2.1's linear equation is $\nabla^4 B = 0$: the Bach vacuum condition says B is *biharmonic*. The forced solution space $\{1/r,\, 1,\, r,\, r^2\}$ is the radial biharmonic kernel, and r^3 and r^{-2}—its nearest neighbours—are verified not to lie in it.

Corollary 2.3 (γr is a point-source response)**.** *For ∇^2 the point-source potential is $1/r$ and its coefficient is the mass; this is Newton, and the origin of $\beta = GM/c^2$. For ∇^4 the corresponding object is r, since $\nabla^2 r = 2/r$ is harmonic away from the origin.*

This is a stronger statement than "γr survives the equation". It is the term the equation *produces* for a localised source, and it sits on the same footing as the Newtonian term rather than being appended to it.

Remark 2.4. That the general static spherically symmetric vacuum solution of conformal gravity has this form is classical. What is contributed here is the removal of the ansatz and the identification of the operator, both computed for an unspecified function.

3 Tully–Fisher forces γ to be universal

With $\beta = GM/c^2$ the locally measured circular velocity is $v^2/c^2 = rB'/2B$, which at small field strength is $\beta/r + \gamma r/2 - kr^2$. In the galactic regime this has a stationary point, and it is a minimum:

$$r_* = \sqrt{2\beta/\gamma}, \qquad v^4 = 2GM\gamma c^2\Big(1 - \tfrac{3}{2}\beta\gamma\Big).$$

Theorem 3.1 (The conditional)**.** *To leading order in $\beta\gamma$: if γ is universal then $v^4 \propto M$, the baryonic Tully–Fisher relation with slope 4 in v. If γ carries a piece proportional to M then $v^4 \propto M^2$. Matching the observed $v^4 = GMa_0$ forces*

$$\gamma \;=\; \frac{a_0}{2c^2}.$$

Remark 3.2 (This agrees with the published normalisation)*.* Conformal gravity fixes the Tully–Fisher coefficient $B/M_\odot = 2c^2G\gamma_0$ [12], i.e. $v^4 = 2c^2G\gamma_0 M$ to leading order, which against $v^4 = GMa_0$ is $\gamma_0 = a_0/2c^2$—the relation above. The identification of $\gamma_0 c^2$ with a universal acceleration of MOND-like magnitude is Mannheim's and is not ours; what the theorem adds is that the baryonic Tully–Fisher relation, taken as a premise, *forces* it rather than merely accommodating it.

Readers comparing numerical factors should note that the same formula gives $a_0 = 2c^2\gamma_0$ at the low-mass end and $4c^2\gamma_0$ where $N^* \sim D$, since $1 + N^*/D \in [1, 2]$. With the fitted $\gamma_0 = 3.06 \times 10^{-30}\,\mathrm{cm}^{-1}$ that brackets 5.5×10^{-9} to $1.1 \times 10^{-8}\,\mathrm{cm\,s}^{-2}$, which is why both factors appear in the literature. Neither is an error.

Read in the reverse-physics direction: the observed Tully–Fisher relation, combined with the forced metric, requires γ to be a universal constant of nature rather than a per-galaxy parameter, and identifies it with the MOND acceleration scale. That is a prediction rather than a fit, and it is falsifiable in a specific way—a mass-dependent γ moves the slope.

Two boundaries travel with it. Identifying $v_{\min}$ with the observed flat velocity is an interpretive step, not a theorem. And the correction factor $1 - \frac{3}{2}\beta\gamma$ is itself mass-dependent, so the slope is exactly 4 only in the limit $\beta\gamma \to 0$; for a galaxy the departure is of order 10^{-14} and unobservable, but "exactly linear" would be too strong.

Remark 3.3. Conformal gravity and MOND are not the same theory and disagree away from r_*: beyond it the conformal curve *rises* ($\mathrm{d}v^2/\mathrm{d}r|_{2r_*} = 3\gamma/8 > 0$) where MOND's are asymptotically flat. Only the $-kr^2$ term can bend it back. The outer rotation curve is therefore a discriminant, not a detail.

4 What may source the theory

§2 and §3 concern the vacuum. γ is a free coefficient there; the explanatory claim needs it *sourced*.

A conformally invariant action satisfies $g_{ab}\,\delta S/\delta g_{ab} = 0$. Applied to the gravitational action this is the Noether identity of the theory; applied to the matter action it reads $T^{\mu}{}_{\mu} = 0$. It is one theorem used twice, and we take it as given.

Theorem 4.1 (Trace obstruction)**.** *A perfect fluid has $T^{\mu}{}_{\mu} = 3p - \rho$, so it is trace-free if and only if $p = \rho/3$. Dust, $p = 0$, forces $\rho = 0$.*

Radiation, uniquely, and pressureless matter not at all. A galaxy's baryons are pressureless to excellent approximation, so on the conformally invariant branch a galaxy is not an admissible source.

The standard repair is to break the symmetry with a conformally coupled scalar acquiring a vacuum expectation value, generating a scale so that ordinary matter can source the theory after all. In its simplest form this fails, and it fails on the term at issue.

Theorem 4.2 (A constant vacuum expectation value forces $\gamma = 0$)**.** *For the conformally coupled scalar, $\Box S + \frac{1}{6}RS + 4\lambda S^3 = 0$ reduces for constant $S = S_0 \neq 0$ to $R = -24\lambda S_0^2$, a constant. For the forced family,*

$$R \;=\; 12k - \frac{6\gamma}{r} + \frac{2(1-w)}{r^2},$$

which is constant in r if and only if $\gamma = 0$ and $w = 1$.

A constant vacuum expectation value does not merely supply a scale; it pins the Ricci scalar, and the forced family is compatible with a pinned R only where the linear term vanishes. Note that u, the Newtonian coefficient, drops out of R entirely: the obstruction is specific to γ.

5 The trichotomy

Collecting §4, the possibilities are organised by one question—*is there a scale, and is it constant or field-dependent?*

	scale	consequence
(A) conformal matter	none	source must be trace-free, hence radiation; a galaxy is excluded, so γ is not tied to M and Tully–Fisher is a coincidence
(B1) constant $\langle S\rangle$	constant	R pinned, hence $\gamma = 0$; the linear term is removed
(B2) non-constant S	field-dependent	γ tied to the scalar profile; §7 makes the tie explicit

(A) does not remove γ; it decouples γ from mass, which destroys the explanation while leaving the term. (B1) removes γ outright. Exactly one branch survives, and it is the one the literature in fact uses.

Corollary 5.1 (Where the assumption lives)**.** *On branch (B2), γ is determined by the scalar profile, which is a property of each configuration. Theorem 3.1 requires γ to be the same constant for every galaxy. Therefore the requirement transfers to the profile.*

That is a question about the symmetry-breaking sector rather than about gravity. §7 carries out the transfer, and it turns out to land on one derived quantity rather than on the profile as a whole.

6 What the conformal gauge costs

Sections 2–5 work in the gauge $g_{tt}g_{rr} = -1$. Two things are needed to justify it.

Proposition 6.1 (Conformal covariance)**.** *In $D = 4$, $B_{ab}[\Omega^2 g] = \Omega^{-2}B_{ab}[g]$, verified for the Mannheim–Kazanas family with symbolic parameters and* unspecified Ω*. Hence Bach flatness is a property of the conformal class, and the classification of §2 classifies classes.*

Proposition 6.2 (Reachability is a first-order ODE)**.** *Setting the new radius to $\rho = \Omega r$ so the angular part remains ρ^2, the gauge condition becomes*

$$r\,\Omega' \;=\; \Omega\big(\Omega\sqrt{ab} - 1\big),$$

first order in Ω, hence locally solvable wherever $ab > 0$ and $r \neq 0$.

What remains assumed is only global continuation—whether Ω stays positive and finite across the exterior. That is strictly smaller than the original assumption and is a question of analysis rather than algebra.

7 Carrying out the transfer

§6 and §2 between them force a third statement that neither makes on its own. Bach flatness is a property of a conformal class; the general solution in the gauge $g_{tt}g_{rr} = -1$ is four parameters subject to one constraint. So the conformal group must *act* on those four parameters.

Theorem 7.1 (The conformal action on the forced family)**.** *Staying in the gauge is not a choice but an ODE, $\mathrm{d}\hat{r}/\mathrm{d}r = (\hat{r}/r)^2$, whose general solution is the Möbius map $\hat{r} = r/(1 - Cr)$—one constant. The induced action is*

$$u \mapsto u, \qquad w \mapsto w + 3uC, \qquad \gamma \mapsto \gamma - 2wC - 3uC^2, \qquad k \mapsto k + C\gamma - C^2 w - C^3 u,$$

a one-parameter group preserving $w^2 + 3u\gamma$ identically. From a member with $\gamma = 0$ it produces $\gamma = -2C - 3uC^2$.

So γ is not an invariant of the Bach vacuum. It is a coordinate on the conformal class, and the classification of §2 classifies classes precisely because of this.

A conformally coupled scalar has weight -1, so the frame in which it is *constant* is reached by exactly this map with $\Omega = S/S_0$—and Theorem 4.2 puts $\gamma = 0$ there. The transfer follows.

Corollary 7.2 (The transfer, in closed form)**.** *On branch (B2) the profile realising the frame change is $S(r) = S_0/(1 - Cr)$ with $C = (S'/S)$ at the origin, and*

$$\gamma \;=\; -2\,(S'/S) \;-\; 3u\,(S'/S)^2 \qquad \textit{so that} \qquad \gamma \simeq -2\,(S'/S).$$

Hence Theorem 3.1's demand that γ be one constant for every galaxy is exactly the demand that S'/S be one constant for every galaxy.

That is more specific than "is the profile universal?". Only one derived quantity has to be universal, and it has to be so across galaxies whose masses differ by orders of magnitude, for a field sourced by the configuration.

Which frame matter follows

Both frames describe the same Bach vacuum, and only one carries a linear term. So the disagreement of [3, 4] is not about the vacuum solution and not about the algebra: it is about which frame matter follows. That premise is itself computable.

Theorem 7.3 (The mass scale is the conformal factor)**.** *A point particle of mass m extremises $-\int m\,\mathrm{d}s$. If the mass is generated by the symmetry-breaking scalar, $m = hS$, then*

$$-\int hS\,\mathrm{d}s_g \;=\; -\int h\sqrt{-S^2 g_{ab}\,\mathrm{d}x^a\mathrm{d}x^b} \;=\; -\int h\,\mathrm{d}s_{S^2 g}$$

identically, so the trajectory is a geodesic of S^2g and not of g. Since S^2g is the frame in which S is constant up to the constant S_0^2, such a particle follows geodesics of the frame with $\gamma = 0$.

Both branches are exhibited with their circular-orbit velocities, and both frames are confirmed Bach vacua, so this is a fork between two admissible solutions of one theory:

mass	trajectory follows	linear coefficient of v^2
m constant, independent of S	geodesics of g	$\gamma/2$
$m = hS$, generated by S	geodesics of S^2g	0

The first entry reproduces Theorem 3.1's weak-field formula by a different route. The disagreement is not an artefact of the expansion: the exact circular-orbit velocities differ too.

A rotation-curve fit needs the first row. The second row is what the symmetry breaking was introduced to provide. Whether a matter sector can generate a mass scale by breaking conformal invariance while leaving particle masses independent of the breaking field is a question about that sector, and we do not answer it. What is no longer available is leaving the choice implicit.

The pair is a solution, and γ is independent of the mass

Theorem 7.1 produces a profile from a transformation law, and a transformation law does not by itself say the result solves anything.

Theorem 7.4 (Admissibility)**.** *In the constant-scalar frame the conformally coupled action collapses to Einstein–Hilbert plus a cosmological term, so its stress tensor is a combination of G_{ab} and g_{ab}. The gravitational equation is $\alpha B_{ab} = \frac{1}{2}T_{ab}$ and that frame's metric is a Bach vacuum, so consistency forces $T_{ab} = 0$—for a constant scalar, the Einstein condition. The metric $1 - u/r - \hat{k}r^2$ is Einstein, with $R = 12\hat{k}$; equating with Theorem 4.2's $R = -24\lambda S_0^2$ gives $\hat{k} = -2\lambda S_0^2$. Bach flatness and the Einstein condition then hold together on the same metric.*

The de Sitter term *is* the scalar potential, and the exterior pair solves the coupled system with nothing left over.

Corollary 7.5 (γ is not a function of the mass)**.** *Theorem 7.1 gives $u \mapsto u$, so the Newtonian coefficient is a conformal invariant of the solution while γ is a coordinate on the class. Two frames with the same u carry different γ. Hence γ requires a second, independent datum—namely C—and for any target γ and any u there is a C attaining it.*

This changes the standing of Theorem 3.1. Requiring γ to be one constant across galaxies of widely differing mass looked like a coincidence in need of explanation; it is not one, because γ cannot depend on mass in any case. What the requirement says is that a single boundary datum on the symmetry-breaking sector is common to every galaxy.

Both readings of that are available and we prefer neither. Favourably, the universality is structurally natural rather than a fine-tuning of gravity. Unfavourably, the explanation of rotation curves is imported from the symmetry-breaking sector rather than produced by gravity.

8 What the source produces

§7 works entirely in the vacuum. Theorem 2.2 identified the governing operator; the multipole integrals it invites were not evaluated there.

Theorem 8.1 (The two moments)**.** *Write the sourced equation as $\nabla^4 B = f$, with f proportional to $T^0{}_0 - T^r{}_r$—the combination Theorem 4.1 already uses. From $\nabla^2 r = 2/r$ the biharmonic Green's function is $-|x-x'|/8\pi$, whose angular average over a shell is $r + r'^2/3r$ outside and $r' + r^2/3r'$ inside. Hence for a compact spherically symmetric source, outside it,*

$$\gamma \;=\; -\tfrac{1}{2}\int f(r')\,r'^2\,\mathrm{d}r', \qquad \beta \;=\; -\tfrac{1}{6}\int f(r')\,r'^4\,\mathrm{d}r'.$$

So γ is the *second* moment of the source—which is the total mass—and β the *fourth*. (We follow the literature's convention, counting powers of r in the radial integrand.) With f tracking the density, $\gamma \propto M$ while the Newtonian coefficient tracks $M\langle r^2\rangle$. For a uniform ball, in closed form, $\gamma = -M/8\pi$ and $\beta/\gamma = R^2/5$; that the ratio is a length squared is what makes "β is not a mass" a computation rather than a reading of the integrand. A shell and a ball of equal mass give the same γ and different β.

This resolves the fork left by Theorem 2.2 on the branch Theorem 3.1 flags: $\gamma \propto M$ gives $v^4 \propto M^2$. It does not end the programme, because Corollary 7.5 supplies the other term.

Remark 8.2 (Which branch this is on). Theorem 8.1 is computed for a perfect-fluid source with no scalar in it, so it lives on the branch where particle masses do *not* track the symmetry-breaking field. On the other branch Theorem 7.3 applies and γ does not act on orbits at all, which makes the moment analysis moot there. The two branches are therefore not alternative calculations of one quantity but calculations in different matter sectors, and we do not reconcile them.

Corollary 8.3 (The decomposition, and a tension we do not resolve)**.** *$\gamma = \gamma_0 + \gamma_*(M)$, with γ_0 a homogeneous datum and γ_* the sourced piece of Theorem 8.1. The superposition is a property of the linear equation $\nabla^4 B = f$; the identification of γ_0 with Corollary 7.5's boundary datum imports the scalar sector, which Theorem 8.1 does not contain.*

That decomposition is not ours: it is the parametrisation the published rotation-curve fits already use, with a universal global term and a local term proportional to the source [6]. And it sits awkwardly with our own Theorem 3.1, which says a mass-proportional piece moves the Tully–Fisher slope, whereas those fits retain both terms and report success over a large sample. We record that as an unresolved tension between a premise we adopt and a fit we do not evaluate. It is the sharpest thing we can say, and it is weaker than a verdict.

Remark 8.4 (A sign worth stating). Theorem 7.4 gives $\hat{k} = -2\lambda S_0^2$, so with $\lambda > 0$ the r^2 coefficient of B is $+2\lambda S_0^2$: the vacuum is anti-de Sitter. A cosmological term of the observed sign would need $\lambda < 0$, and then the scalar potential is unbounded below. We note the fork; we do not take it.

Separately, the fourth-moment result for β is the substance of the standing objection to conformal gravity's Newtonian limit [3, 5], arrived at here from the source side. We exhibit it; we do not argue from it.

9 What this paper does not establish

- **No novelty in the physics, anywhere.** The Mannheim–Kazanas solution, its reduction to a fourth-order Poisson problem, and the trace-free condition on conformal matter are classical. So is the rest. Theorem 8.1 is Mannheim and Kazanas' [5] and is the substance of the objection to the Newtonian limit. Theorem 7.3 is Horne's [7], stated there to the level of the formula—rest masses scale with $S(r)/S_0$, so world lines follow geodesics of $\Omega^2 g$ with $\Omega = S/S_0$, which is also Theorem 7.1's conformal factor. Theorems 7.3 and 7.4 are then Hobson and Lasenby's [10]: that timelike geodesics are not conformally invariant, that the scalar generating particle mass puts massive particles on geodesics of the Schwarzschild–de Sitter metric, and that the resulting curves have no flat region—a *stronger* conclusion than Remark 8.2, which leaves the branch open. That massive particles follow the frame in which their mass is constant is standard scalar–tensor material. For Theorem 7.1 we have not found the group action on (w, u, γ, k) written in this form, which is not a claim that it is new. The contribution of this paper is that each statement is computed with the ansatz removed, the premises isolated, and controls that fail when a claim is vacuous—not the statements themselves.
- **No fits.** No galaxy data enters anywhere. The only observational inputs are the *shape* of the Tully–Fisher relation, used as a premise in Theorem 3.1, and that galaxies are pressureless, used in §4. Neither is our measurement.
- **No interior solution, and no numbers.** Theorem 7.4 closes the exterior and Theorem 8.1 gives the exterior coefficients as functionals of the source, but the interior field and the scalar's profile inside matter are not solved. Nor is it shown that a cosmological boundary condition *can* deliver a universal γ_0. Above all, Corollary 8.3 is not evaluated: whether $\gamma_* \ll \gamma_0$ holds for real galaxies is observational and we do not measure it. The assumption is named and sharpened, not discharged.
- **Two matter descriptions, not reconciled.** Theorem 7.3 takes a mass generated by a Yukawa coupling, $m = hS$; Theorems 4.1 and 8.1 use a perfect fluid. Both are standard and both are entered as premises. Whether they are compatible is a matter-sector question we do not address.
- **No resolution of the dispute, and no prospect of one by these methods.** The exchange is live and long: Horne [7], then Hobson and Lasenby [10, 9], against Mannheim [4, 8], over more than a decade, with no consensus. Theorem 7.3 narrows it to one premise and computes both branches, which is all we do. We stress that the crux is *not algebraic*: both sides agree on the transformation law. They disagree over whether the mass-generating scalar is macroscopic or microscopic—whether a vacuum expectation value varies across a galaxy or only within particle interiors [8]. That is a physical premise, and exact symbolic computation with controls takes such premises as input. It cannot settle this one.
- **Spherical symmetry, for a claim about discs.** Everything is static and spherically symmetric. Galaxies are not, and the rotation-curve literature works the disc case for that reason. The gauge assumption is discharged as far as it can be in §6; the geometry assumption is not discharged at all.

- **Nothing about the ghost.** Conformal gravity's fourth-order kinetic operator carries an Ostrogradsky ghost. That is orthogonal to everything here and is treated elsewhere in this programme.
- **No numerical value.** a_0 is not measured here and no published γ is compared; $\gamma = a_0/2c^2$ is an algebraic consequence of a premise, not a numerical agreement.

10 Verification

Each result is a fail-closed certificate with a deterministic verifier. Controls are designed to fail if the claim were vacuous; several did so during development and are recorded in the certificates.

result	certificate	checks
Thm. 2.1	`BHOB_GENERAL_STATIC_SPHERICAL_COMPLETENESS`	24
Thm. 3.1	`BHOC_TULLY_FISHER_SCALING`	14
Thm. 2.2	`BHOD_BACH_VACUUM_IS_BIHARMONIC`	9
Thm. 4.1	`BHOE_MATTER_TRACE_DILEMMA`	8
Thm. 4.2	`BHOF_CONSTANT_VEV_FORCES_GAMMA_ZERO`	12
§6	`BHOG_CONFORMAL_CLASS_INVARIANCE`	10
Thm. 7.1	`BHOH_UNIVERSALITY_TRANSFER`	18
Thm. 7.3	`BHOI_WHICH_FRAME_MATTER_FOLLOWS`	13
Thm. 7.4	`BHOJ_THE_ASSUMPTION_NAMED`	12
Thm. 8.1	`BHOK_THE_SOURCE_MOMENTS`	22

Theorems 2.1, 4.2, 7.1, 7.3, 7.4 and 8.1, and §6, are computed in SYMPY; Theorems 2.2 and 4.1 are computed in Forge over exact rationals with symbolic differentiation of unspecified functions. Theorem 2.1's conclusion carries an independent Forge rail using a different arithmetic stack, representation and code path; that rail confirms which metrics are Bach-flat but not the row decomposition itself, which remains single-implementation.

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